

## LITERATURE SURVEY

| Paper                                                                                                                                                                        | Methodology                | Dataset                | Contributions                                                                       | Limitations                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>A Deep Learning Approach for Helmet Detection and Fine Generation System (2025)</b>                                                                                       | YOLOv3 ,OCR                | Kaggle/Custom Dataset  | Helmet Detection, Number Plate Recognition, Fine Generation                         | No Triple Riding Detection, No Rider Tracking      |
| <b>Computer-Vision Based Automatic Rider Helmet Violation Detection and Vehicle Identification in Indian Smart City Scenarios Using NVIDIA TAO Toolkit and YOLOv8 (2025)</b> | DetectNet_v2, YOLOv8 , OCR | Indian Traffic Dataset | Rider Detection, Helmet Detection, Vehicle Identification, Number Plate Recognition | No Automated Challan Generation                    |
| <b>Automatic Two-Wheeler Rider Identification and Triple-Riding Detection in Surveillance Systems Using Deep Learning Models (2025)</b>                                      | DetectNet_v2 , YOLOv8      | Custom Traffic Dataset | Rider Detection, Triple Riding Detection                                            | No Number Plate Recognition, No Challan Generation |

| Paper                                                                                                                                             | Methodology                                                                                                                                    | Dataset                                                                     | Contributions                                                                                                                   | Limitations                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Automatic Helmet Detection from Videos Using Artificial Intelligence Method and Engage with Secured Driving System by Message Alert (2025)</b> | Adaptive YOLOv5 , ViT-DRDNet for helmet detection                                                                                              | Public YouTube traffic videos converted into image frames                   | Motorcycle detection, Helmet detection, Number plate recognition for violators, Automatic SMS/message alert to registered owner | Does not detect triple riding/passenger violations, No automatic e-challan generation             |
| <b>A Safety Detection Method for Electric Bike Riding Incorporating License Plate Region Recognition (2025)</b>                                   | F-YOLOv8 (Improved YOLOv8) integrated with Feature Pyramid Network (FPN), Transformer attention mechanism and license plate region recognition | Custom Electric Bike Dataset                                                | Rider Detection, Helmet Detection, Vehicle Identification, Number Plate Recognition                                             | No Automated Challan Generation, Focused only on electric bikes                                   |
| <b>Improved YOLOv8n Model for Detecting Helmets and License Plates on Electric Bicycles (2024)</b>                                                | YOLOv8n ,Weighted Concat, lightweight GCC convolution, OCR for license plate recognition                                                       | Custom electric bicycle dataset collected from real traffic scenes in China | Rider Detection, Number Plate Recognition                                                                                       | Focuses only on electric bicycles, Does not detect triple riding, No automatic challan generation |